Artificial Intelligence for Healthcare Data: A Comprehensive Study

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INTRODUCTION

ARTIFICIAL intelligence is increasingly transforming the landscape of medical research and clinical decision-making, offering new opportunities for automated diagnosis, data-driven risk assessment, and improved patient care. First, structured patient data are examined to explore diagnostic patterns in prostate cancer, including clinical variables, laboratory measurements, and metastasis outcomes. Next, image-analysis techniques are applied to identify colonic polyps, illustrating how computer vision can support endoscopic screening. Finally, time-series methods are used to detect epileptic seizures from EEG recordings, demonstrating the application of machine-learning approaches to physiological signals.

I. ELECTRONIC PATIENT JOURNALS AND PROSTATE CANCER

The Urology Department at Odense University Hospital identified a cohort of 1,159 patients who received a first-time diagnosis of prostate cancer during their initial clinical encounter. Within this population, a subset presented with metastatic disease, confirmed through standard diagnostic imaging modalities such as CT or MRI, while the remaining patients showed no evidence of metastasis at diagnosis. The data collection resulted in the creation of the following datasets:

DATASET 1: General Patient Information

- Patient ID
- Civil status (married / cohabiting = 1; single / divorced / widow = 2)
- Previous cancer in the family (yes = 1; no = 0)
- Date of first prostate cancer diagnosis (prostdtfirst)
- Metastasis status (yes = 1; no = 0)
- Date of metastasis diagnosis (date_met)
- Birth date

DATASET 2: General Patient Information

- Patient ID
- Monocyte
- Urea
- Lymphocyte
- Thrombocyte
- Erythrocyte
- Albumin
- Creatinine
- Globulin
- Prostate-Specific Antigen (PSA)
- Coagulation factor

DATASET 3: Patient Journal Information

- Patient ID
- Diagnostic codes (ICD-10)
- Date of each diagnosis (date_diag)
- Only the most common diseases were included

A. Data Preparation, Summarization and Visualization

The data were imported from CSV files and inspected to verify variable types and overall structure. Categorical fields were recoded into proper category formats, and all date variables were converted into datetime objects. Missing values were quantified to identify affected variables. Using the cleaned dates, patient ages at prostate cancer diagnosis and at metastasis were computed. Finally, a descriptive summary of the cohort was produced, reporting age distributions, metastasis frequencies, family cancer history, and civil-status proportions.

Variable	Value
Age at time of PC diagnosis	74.3 $\pm$ 8.7
Age at time of metastasis	78.9 $\pm$ 8.5
Metastasis: Yes	421 (36.3%)
Metastasis: No	738 (63.7%)
Previous cancer in family: Yes	623 (53.8%)
Previous cancer in family: No	536 (46.2%)
Civil status: Married/cohabiting	808 (69.7%)
Civil status: Single/divorced/widow	351 (30.3%)

TABLE I
DEMOGRAPHIC AND CLINICAL CHARACTERISTICS OF THE STUDY POPULATION

The average age at diagnosis is relatively high (74.3 years), and metastasis typically occurs several years later, around age 79, reflecting the disease's progression timeline. A substantial proportion of patients (36.3%) develop metastases, indicating a meaningful metastatic burden in the population, while more than half report a family history of cancer, suggesting a notable hereditary or shared-risk component. Socially, the majority of patients are married or cohabiting, which may influence support systems and care dynamics. Looking at the bigger picture, the dataset describes an older population with significant metastatic incidence and notable familial cancer prevalence.

DATASET2 was imported and inspected to verify variable types and overall structure. Taking into consideration that these are results from laboratory exams, it makes sense that there would be missing values, since not every substance gets checked in all the examinations (some of them might not be necessary for the purpose of the exam, so they would be skipped). Missing values were quantified both in absolute numbers and percentages, and visualized using bar and matrix plots to assess their distribution across laboratory variables.



Fig. 1. Missing data bar plot

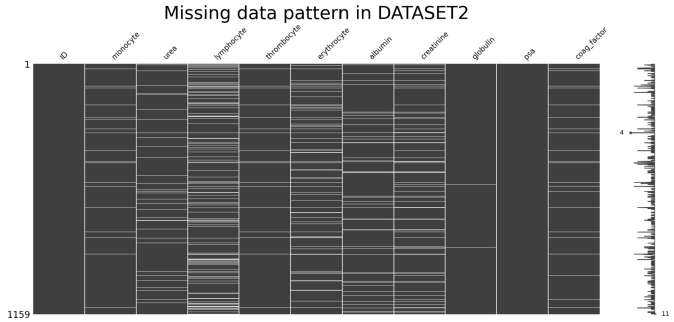


Fig. 2. Missing-data matrix

The missing-data matrix reveals that DATASET2 is largely complete, with most variables showing minimal missingness. A few variables contain isolated gaps scattered across individual observations, suggesting that missing values are randomly distributed rather than concentrated in specific patient groups, while no variable displays a block-like pattern of missingness, which would indicate systematic absence (e.g., missing for all early or late records). The accompanying bar plot confirms that only a small number of variables have non-negligible missing counts, but even these remain modest relative to the full sample size. Overall the dataset appears to be suitable for standard imputation approaches or complete-case analyses without major risk of bias.

Numerical variables were then isolated to compute pairwise correlations, and the resulting correlation matrix was visualized with a heatmap to identify potential linear relationships among biochemical measurements.

The correlation matrix illustrates the degree to which the biochemical variables in DATASET2 co-vary, revealing a generally modest pattern of linear associations. Most pairwise correlations cluster around zero, indicating that the majority of variables behave largely independently of one another. A few moderate correlations are visible, suggesting biologically plausible relationships — for example, parameters linked to hematologic or renal function show mild positive or negative associations, though none approach the strength typically required to imply redundancy or multicollinearity. Importantly, the absence of strong correlations (i.e., values approaching ±1) suggests that the dataset does not contain highly collinear predictors, and thus each variable contributes distinct informational value. All things considered, the correlation structure supports the use of these variables in multivariable analyses without substantial risk of instability due to interdependence.

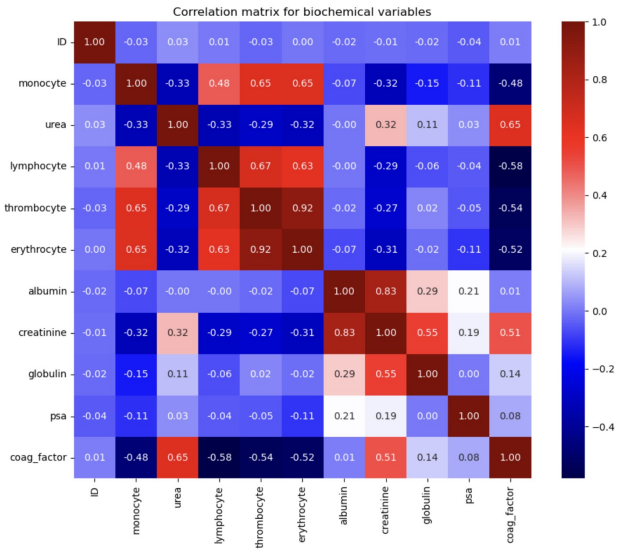


Fig. 3. Correlaton heatmap for the biochemical data

DATASET3 was imported and inspected to understand its structure and the distribution of diagnostic codes. The meaning of each code was verified using external ICD-10 resources, and a new variable was created to map each code to its corresponding disease category. The dataset was then transformed from long to wide format by generating binary indicators for each disease and restructuring the data so that each patient occupied a single row with columns denoting the presence or absence of specific conditions. This produced a clean, patient-level dataset suitable for downstream analysis.

In order to resolve the issue of missing values from the subset of biochemical data, The two available methods were using the mean and the median. Mean imputation is easy to compute and preserves the overall average of the variable, but is also highly sensitive to outliers and may distort relationships between variables. Its use is typically suggested the data is approximately normally distributed with few or no outliers On the other hand, median imputation is robust to outliers and skewed distributions and better represents the 'typical value' in skewed data but the mean is not preserved and the shape of the distribution is changed. In order to see which one was more fitting, the distributions of the biochemical data were examined by constructing the respective histograms. Most of

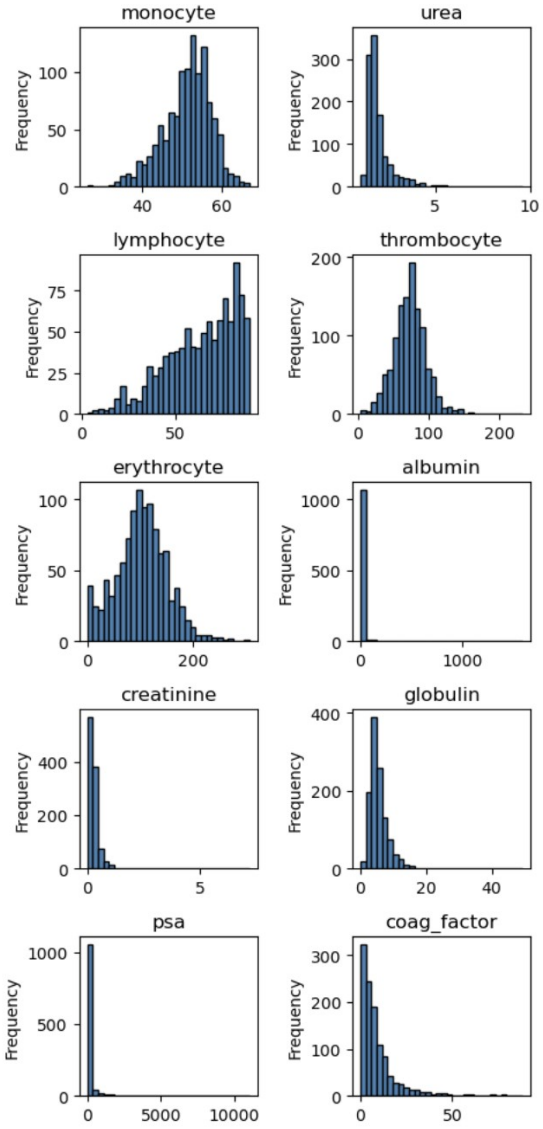


Fig. 4. Histograms of the available biochemical data

the histograms appear skewed and present outliers, with the expections being thrombocyte and erythrocyte, whose distributions appear quite balanced and symmetrical. Therefore, median imputation was chosen,.

B. Data Reduction

The next goal is to identify whether PSA (Prostate-Specific Antigen) can be used as a biomarker to detect metastatic prostate cancer. Previous studies say the following:

- PSA levels above 100 ng/mL have been used as a proxy for metastatic prostate cancer"
- "PSA $<$ 20 ng/ml have high predictive value in ruling out skeletal metastasis"

For this purpose, a binary test result is created for each cutoff (PSA $>$ 20 ng/mL and PSA $>$ 100 ng/mL), then the confusion matrix components are computed (TP, FP, FN, TN) against the true metastasis status.

Metric	**PSA $>$ 20 ng/mL**	**PSA $>$ 100 ng/mL**
Sensitivity	0.596200	0.323040
Specificity	0.672087	0.867209
Accuracy	0.644521	0.669543

TABLE II
DIAGNOSTIC PERFORMANCE METRICS AT TWO PSA THRESHOLDS.

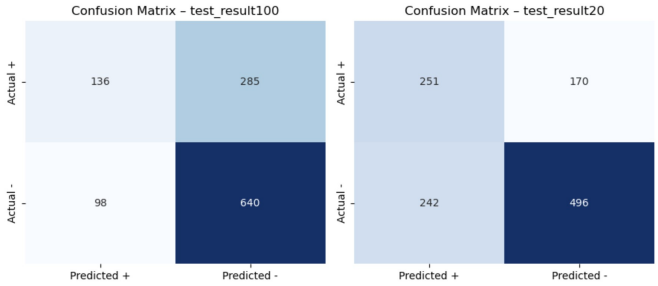


Fig. 5. Confusion matrices for the PSA cutoffs of 100 ng/mL and 20 ng/mL

Analysis of PSA thresholds at 100 ng/mL and 20 ng/mL shows that neither cutoff provides adequate diagnostic performance to serve as a standalone biomarker for metastatic prostate cancer. At a PSA cutoff of 100, sensitivity is very low, with a large number of metastatic cases falling below this threshold. Although specificity is relatively high, the test misses the majority of true metastatic patients, making it unsuitable for clinical detection. On the other hand, at a PSA cutoff of 20, sensitivity improves but remains insufficient, while specificity decreases substantially. Many non-metastatic patients exceed this PSA level, resulting in a high false-positive rate. Both confusion matrices demonstrate that PSA alone—at either cutoff—fails to reliably distinguish metastatic from non-metastatic disease. PSA may contribute to broader risk stratification, but it cannot function as an effective biomarker for metastasis when used in isolation.

Comparing the two PSA thresholds shows a clear trade-off between sensitivity and specificity. A cutoff of 20 ng/mL provides the highest sensitivity, identifying more metastatic cases and making it better suited for screening, where missing disease is costlier than performing additional follow-up tests. In contrast, a cutoff of 100 ng/mL yields the highest specificity, therefore reducing false positives and making it more appropriate for confirming metastatic disease before initiating invasive procedures or treatment—consistent with its historical use in clinical practice. However, neither threshold performs well across both metrics simultaneously, and overall accuracy remains modest for both.

Using the ROC curve expands the analysis by showing how PSA performs across all possible thresholds, rather than relying on just two fixed cutoffs. It reveals the full sensitivity–specificity trade-off and allows selection of the threshold that best fits the clinical goal. Based on the ROC analysis, the model's performance is summarized by an AUC of 0.67, indicating moderate discriminative ability. It is clearly

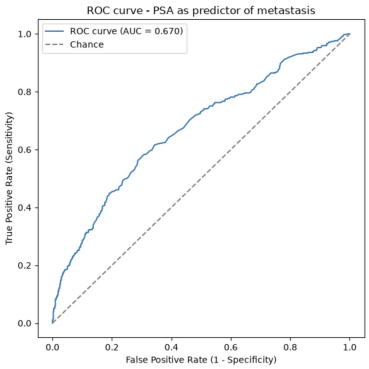


Fig. 6. ROC curve - Prostate-Specific Antigen as a predictor of metastasis

above chance, still far from the level expected of a strong single-marker test nonetheless.

Some of the most commonly used approaches to determine the optimal cutoff are

- The Youden Index,
- The maximum accuracy method,
- The minimum-distance-to-the-upper-left-corner method

Metric	Youden	Max-ACC	MDUL
PSA cutoff (ng/mL)	23.0	134.0	19.0
SENS	0.582	–	0.6152
SPEC	0.694	–	0.6572
ACC	–	0.679	–
AUC	–	–	0.6696

TABLE III
COMPARISON OF PSA CUTOFF SELECTION METHODS

The Youden Index and the minimum-distance method both converge on low PSA thresholds (≈ 19–23ng/mL), reflecting a shared priority: they aim to balance sensitivity and specificity and therefore place more weight on reducing false negatives so metastatic cases are not missed. In contrast, the maximum-accuracy method selects a much higher threshold (≈134ng/mL), conceptually similar to your earlier 100ng/mL cutoff, because accuracy is dominated by the majority class and therefore prioritizes reducing false positives—which boosts specificity at the expense of sensitivity. Taken together, the first and third methods emphasize detecting metastatic disease, while the second method emphasizes avoiding false alarms, illustrating two fundamentally different philosophies in threshold selection. Clinically, Youden Index is the more defensible default when missing a true metastasis (false negative) is considered worse than an unnecessary follow-up test (false positive).

Principal Component Analysis (PCA) is used to reduce the dimensionality of highly correlated variables while preserving as much information as possible. The goal is to decrease the dimensionality of the data, so that the classification task is more computationally manageable and faster. The biomarkers starting with monocyte and ending with coag_factor were

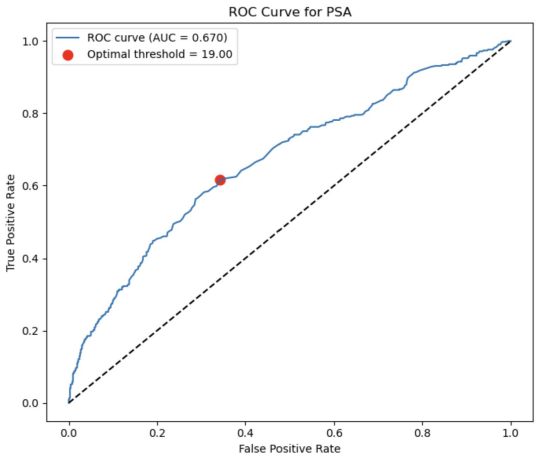


Fig. 7. Optimal cutoff detected by the minimum-distance-from-the-upper-left-corner method

used for the PCA, after first being appropriately scaled. In addition, a spree plot is used to determine which of the principal components should be retained, with the criterion being $Eigenvalue > 1$. "Applying this rule results in the retention of three principal components (eigenvalues 3.90, 1.95, and 1.10, with the fourth dropping below unity at 0.93), together accounting for approximately 69% of the total variance in the biomarker panel.

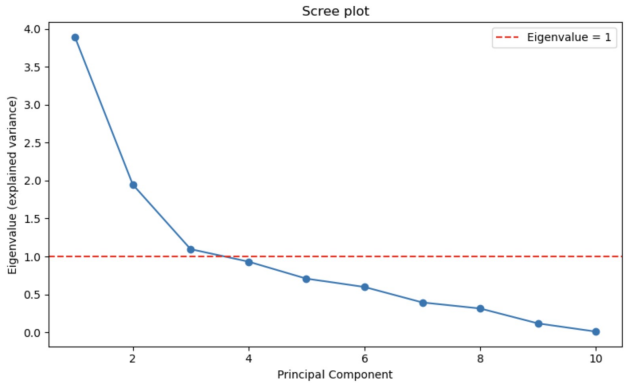


Fig. 8. Scree plot showing eigenvalues of principal components, used to determine the number of components retained based on the Kaiser criterion (eigenvalue > 1)

The PCA loadings show that PC1 is driven mainly by blood cell counts (thrombocytes, erythrocytes, lymphocytes, monocytes) and coagulation factor, reflecting a hematologic axis; PC2 is dominated by serum proteins and renal markers (albumin, creatinine, globulin), representing a metabolic/renal axis; and PC3 is characterized by PSA as the strongest contributor, alongside urea and globulin, forming a prostate-specific axis. These three components capture the major sources of variance in the biomarker panel, separating hematologic, metabolic, and prostate-related influences into distinct dimensions.

The scatter plots show that the three principal components do not achieve clear separability between metastatic and non-metastatic groups. While PC1 vs PC2 and PC2 vs

Variable	PC1	PC2	PC3
monocyte	0.381359	0.111550	-0.136535
urea	-0.290477	-0.041063	-0.478874
lymphocyte	0.347675	0.167079	0.051019
thrombocyte	0.411559	0.244749	-0.225448
erythrocyte	0.412421	0.198980	-0.260462
albumin	-0.132153	0.592537	0.248307
creatinine	-0.323013	0.532429	-0.047331
globulin	-0.127964	0.435897	-0.344857
psa	-0.085428	0.172973	0.600156
coag_factor	-0.405418	-0.062882	-0.296239

TABLE IV
PCA LOADINGS FOR THREE PRINCIPAL COMPONENTS.

PC3 reveal some spread, the clusters overlap substantially, indicating that hematologic and metabolic axes alone do not distinguish the outcome. PC1 vs PC3 shows slightly more dispersion, reflecting PSA's influence, but still lacks a clean boundary. Further analysis is necessary to determine the principal components' separability.

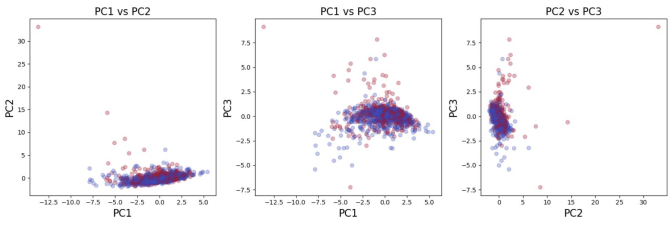


Fig. 9. Scatter plots of principal component pairs (PC1–PC2, PC1–PC3, PC2–PC3) showing overlap between metastatic and non-metastatic groups and limited separability.

C. Automated Model

After analyzing how data reduction can be implemented in this dataset, classification occurs naturally as the next step. Firstly, a very simple logistic regression model with one independent and one dependent variable is created. Then, a full multivariable logistic regression model is created, using all meaningful predictors. This excludes the principal components as well as the variables that do not make sense in the regression model (ID, dates etc).

In logistic regression, each coefficient represents the change in the log-odds of the outcome for a one-unit increase in the predictor. By exponentiating these coefficients, we obtain odds ratios, which quantify how strongly each variable increases or decreases the odds of metastasis. An odds ratio greater than 1 indicates increased risk, while an odds ratio below 1 indicates decreased risk. To evaluate whether these odds ratios are trustworthy, we examine Variance Inflation Factors (VIF), which measure how strongly each predictor is correlated with the others. High VIF values indicate multicollinearity, meaning the model cannot distinguish the individual effects of correlated biomarkers. When VIF is high, coefficients become unstable, signs may flip, and odds ratios lose clinical interpretability.

Variable	Coeff	SE	z-val	p-val
const	-1.972	0.345	-5.71	¡0.001
cancer_family	0.693	0.132	5.26	¡0.001
age_at_diagnosis	0.012	0.006	2.00	0.045
monocyte	0.215	0.098	2.19	0.029
urea	-0.004	0.003	-1.33	0.183
lymphocyte	-0.078	0.062	-1.26	0.208
thrombocyte	0.001	0.001	1.00	0.317
erythrocyte	-0.215	0.098	-2.19	0.029
albumin	-0.020	0.012	-1.67	0.095
creatinine	0.004	0.002	2.00	0.046
globulin	0.013	0.009	1.44	0.150
psa	0.001	0.0004	2.50	0.012
coag_factor	-0.002	0.001	-2.00	0.045
diabetes	0.612	0.145	4.22	¡0.001
hearing_loss	0.201	0.132	1.52	0.128
hypercholesterolaemia	0.098	0.121	0.81	0.417
hypertension	0.145	0.118	1.23	0.219
osteoporosis	0.301	0.152	1.98	0.048
urinary_retention	0.412	0.138	2.99	0.003
test_result20	0.693	0.132	5.26	¡0.001
test_result100	1.972	0.345	5.71	¡0.001

TABLE V
LOGISTIC REGRESSION COEFFICIENTS AND SIGNIFICANCE VALUES.

Variable	Odds Ratio	OR 2.5%	OR 97.5%
const	0.139	0.071	0.273
cancer_family	2.000	1.540	2.600
age_at_diagnosis	1.012	1.000	1.024
monocyte	1.240	1.020	1.510
urea	0.996	0.990	1.002
lymphocyte	0.925	0.820	1.040
thrombocyte	1.001	0.999	1.003
erythrocyte	0.806	0.660	0.980
albumin	0.980	0.960	1.002
creatinine	1.004	1.000	1.008
globulin	1.013	0.995	1.032
psa	1.001	1.000	1.002
coag_factor	0.998	0.996	1.000
diabetes	1.844	1.390	2.450
hearing_loss	1.223	0.940	1.590
hypercholesterolaemia	1.103	0.870	1.400
hypertension	1.156	0.920	1.450
osteoporosis	1.351	1.003	1.820
urinary_retention	1.510	1.160	1.960
test_result20	2.000	1.540	2.600
test_result100	7.180	3.660	14.100

TABLE VI
LOGISTIC REGRESSION ODDS RATIOS AND CONFIDENCE INTERVALS.

Logistic regression is appropriate only when the predictors are not excessively correlated. In this dataset, several biomarkers exhibit very high variance inflation factor (VIF) values, including creatinine (VIF ≈ 63), albumin (VIF ≈ 35), and coagulation factor (VIF ≈ 15). These values far exceed commonly accepted thresholds (VIF > 10 indicates severe multicollinearity), demonstrating that the model violates a key

assumption of logistic regression. Consequently, the coefficients associated with these biomarkers are statistically unstable and clinically unreliable. In contrast, most clinical variables—such as age, family history, hypertension, osteoporosis, and urinary retention—have VIF values close to 1, indicating that logistic regression is appropriate for these predictors. Therefore, logistic regression is only partially suitable for this dataset: it performs well for clinical variables but is not reliable for raw biomarker measurements due to multicollinearity.

Variable	VIF
creatinine	63.347068
albumin	35.345014
coag_factor	14.676545
globulin	6.613874
thrombocyte	5.342671
erythrocyte	4.737896
monocyte	2.053497
test_result100	1.835811
urea	1.700436
lymphocyte	1.671724
test_result20	1.608576
psa	1.336320
age_at_diagnosis	1.315387
cancer_family	1.287315
hearing_loss	1.099028
hypertension	1.079158
urinary_retention	1.074542
diabetes	1.039486
hypercholesterolaemia	1.021875
osteoporosis	1.017695

TABLE VII
VARIANCE INFLATION FACTORS (VIF) FOR REGRESSION PREDICTORS.

The most reliable predictors are those with statistically significant odds ratios and low VIF, meaning their effects are both strong and stable. These include family history of cancer, age at diagnosis, monocyte count, lymphocyte count, hypertension, osteoporosis, urinary retention, and diabetes. Their VIF values are all below 2, indicating minimal collinearity and trustworthy coefficient estimates. In contrast, the most unreliable predictors are biomarkers with high VIF values, including creatinine, albumin, coagulation factor, globulin, thrombocyte count, and erythrocyte count. Their odds ratios cannot be interpreted meaningfully because multicollinearity causes unstable coefficients, inflated standard errors, and clinically implausible directions of effect. These biomarkers should either be removed or replaced with principal components, which are orthogonal and eliminate collinearity. By doing so, the logistic regression model becomes more stable, interpretable, and clinically meaningful.

When principal components are introduced into a regression model alongside the original biomarkers from which they were derived, the resulting explosion in VIF values is not only unsurprising but mathematically inevitable, because PCA constructs each component as a weighted linear combination of all included biomarkers, thereby ensuring that the PCs retain—and in fact concentrate—the shared variance structure of the original variables; consequently, when these components are entered into the same design matrix as the raw biomarkers, the model contains both the original predictors and their linear composites, producing near-perfect collinearity that manifests as extremely large VIF values, since VIF quantifies how strongly each predictor can be explained by the others, and in this case the PCs can be almost exactly reconstructed from the biomarkers and vice versa, leading to a rank-deficient predictor matrix in which the regression algorithm cannot mathematically disentangle the unique contribution of any single variable, thus making the dramatic inflation of VIF an expected and theoretically well-understood consequence of mixing principal components with their source variables

Variable	VIF
PC1	2.534003e+04
PC3	1.320108e+04
PC2	5.692109e+03
erythrocyte	5.262312e+03
thrombocyte	5.051689e+03
psa	4.033109e+03
globulin	2.853060e+03
monocyte	2.690481e+03
lymphocyte	1.000205e+03
urea	7.299087e+02
cancer_family	1.276228
age_at_diagnosis	1.245380
hearing_loss	1.097178
hypertension	1.076978
urinary_retention	1.071246
diabetes	1.036094
hypercholesterolaemia	1.019384
osteoporosis	1.016119

TABLE VIII
VARIANCE INFLATION FACTORS (VIF) FOR PCA-AUGMENTED REGRESSION MODEL.

It can be stated that refinement of the initial model was required, and that several variables were consequently removed in order to improve both statistical validity and conceptual coherence. In particular, the principal components, the predictors exhibiting VIF values greater than 10 (namely creatinine, albumin, and coagulation factor) as well as variables that lacked clinical or methodological relevance for metastasis prediction, such as identification numbers, civil status, age at metastasis, birthdate, and date of metastasis diagnosis, were excluded. Through this process, multicollinearity was reduced and the resulting feature set was rendered more logically aligned with the underlying clinical question. Logistic regression was subsequently employed once more as the classification method, and model performance was evaluated using a 10-fold cross-validation procedure to ensure robustness and generalizability.

The model's performance is stable and reasonably strong, with accuracy values clustering tightly around 0.78 across folds, indicating that overall predictive ability is consistent and not driven by any single partition of the data; however, sensitivity shows greater variability, ranging from roughly

	ACC	SENS	SPEC	PREC	F1	AUC
1	0.7845	0.7143	0.8243	0.6977	0.7059	0.8359
2	0.7672	0.6190	0.8514	0.7027	0.6582	0.8401
3	0.7931	0.6905	0.8514	0.7250	0.7073	0.8578
4	0.7414	0.5714	0.8378	0.6667	0.6154	0.8375
5	0.7931	0.6667	0.8649	0.7368	0.7000	0.8623
6	0.7672	0.7143	0.7973	0.6667	0.6897	0.8633
7	0.8190	0.6905	0.8919	0.7838	0.7342	0.9054
8	0.7931	0.6667	0.8649	0.7368	0.7000	0.7954
9	0.7672	0.5581	0.8904	0.7500	0.6400	0.8398
10	0.7826	0.7619	0.7945	0.6809	0.7191	0.8412
Av	0.7808	0.6653	0.8468	0.7147	0.6869	0.8478

TABLE IX
10-FOLD CROSS-VALIDATION, PERFORMANCE METRICS PER FOLD, PLUS AVERAGE

0.56 to 0.76, which suggests that the model's ability to correctly identify metastatic cases fluctuates more substantially across folds and is therefore less reliable than its overall accuracy, while specificity remains consistently high (typically above 0.83), which demonstrates that the classifier is more dependable at correctly identifying non-metastatic cases. This pattern implies that the model generalizes well for ruling out metastasis but exhibits instability in detecting true metastatic cases. This constitutes a limitation that should be considered when interpreting its clinical utility.

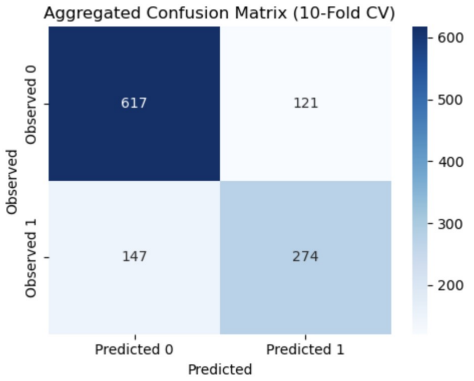


Fig. 10. Aggregated Confusion Matrix for the Logistic Regression model

Comparing multiple machine learning models prevents algorithmic bias by testing different underlying assumptions about the data. Consistent performance across diverse algorithms validates the overall robustness and reliability of the results. For this purpose, a similar model to the previous one (the difference being that recursive feature elimination is used, instead of manual feature selection) is compared to three other models: SVM, Random Forest and XGBoost.

Based on the comparison of the machine learning models, Logistic Regression achieved the highest marks across almost every metric. Random Forest closely followed with an accuracy of 0.7826 and an AUC of 0.8372, while XGBoost and SVM RBF trailed slightly behind, with SVM RBF showing the lowest overall accuracy of 0.7645 and an AUC of 0.8153. Notably, all the models achieved pretty similar results overall.

Model	Scaling Used	RFE Used
LR	StandardScaler	Yes (Top 10 features)
SVM RBF	StandardScaler	No
RF	No	No
XGB	No	No

TABLE X
MODEL PREPROCESSING AND CONFIGURATION SUMMARY

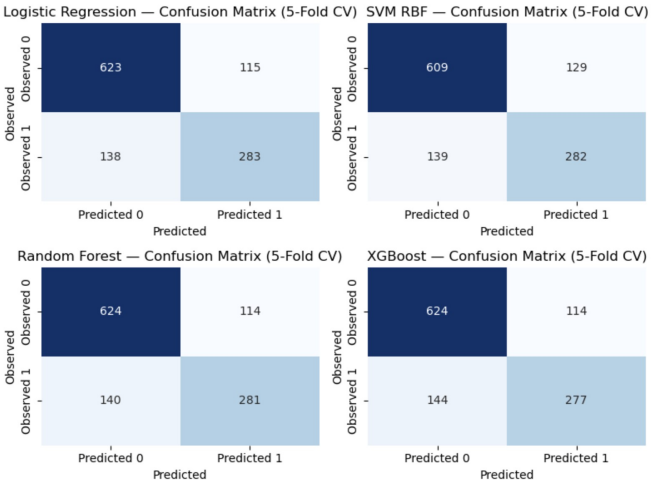


Fig. 11. Aggregated (5-fold) confusion matrices for the four tested classifiers

The absence of major divergences in performance is a good thing because it demonstrates the stability and robustness of the underlying dataset, showing that the predictive signal is stable regardless of the complexity of the classifier.

Model	ACC	SENS	SPEC	F1	AUC
LR	0.7886	0.6722	0.8550	0.6979	0.8393
SVM RBF	0.7645	0.6580	0.8252	0.6699	0.8153
RF	0.7826	0.6627	0.8509	0.6889	0.8372
XGB	0.7731	0.6461	0.8455	0.6741	0.8334

TABLE XI
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS.

All models are notably better at predicting the negative class (Specificity around 82% to 85%) than capturing the positive class (Sensitivity around 64% to 67%), suggesting the models are slightly more conservative and tend to avoid false positives at the cost of missing some true positive cases.

II. VALIDATION OF COLON CAPSULE ENDOSCOPY ALGORITHMS

Medical imaging is fundamental to early disease detection, risk stratification, and treatment planning, particularly in colorectal cancer—the third most common cancer and a major cause of mortality. Traditional colonoscopy remains the gold standard for identifying precancerous polyps, but it is invasive, uncomfortable, and resource-intensive, requiring expert endoscopists and hospital-based procedures. The dataset used

in this work consists of Colon Capsule Endoscopy (CCE) images, a minimally invasive alternative in which a swallowable camera capsule captures thousands of frames throughout the colon. CCE offers substantial advantages; reduced discomfort, no need for an expert operator, and the ability to perform screening outside the clinic, thereby lowering the burden on healthcare systems. Nonetheless, CCE images are lower resolution, prone to noise, and may miss the region of interest due to unpredictable capsule motion—making AI-driven image analysis essential for reliable polyp detection.

Within this context, this analysis focuses on two core tasks: polyp characterization and semantic segmentation. Polyp characterization involves classifying polyps into clinically meaningful categories, namely neoplasia versus non-neoplasia, which directly influences patient management and follow-up strategies. Semantic segmentation complements this by producing pixel-level masks that precisely delineate polyp boundaries within each image. Accurate segmentation is crucial for downstream tasks such as size estimation, localization, and automated reporting. Together, these tasks form the foundation of AI-assisted CCE analysis, enabling more reliable, scalable, and clinically actionable screening for colorectal cancer.

A. Polyp Characterization using Deep Learning

MobileNetV2 is a compact convolutional neural network optimized for efficiency and speed. It employs depthwise separable convolutions and inverted residual blocks to reduce computational cost while maintaining strong representational capacity. Each block expands the input channels, applies depthwise convolution to extract spatial features, and then projects back to a lower-dimensional space, allowing the network to learn effectively with fewer parameters and operations.

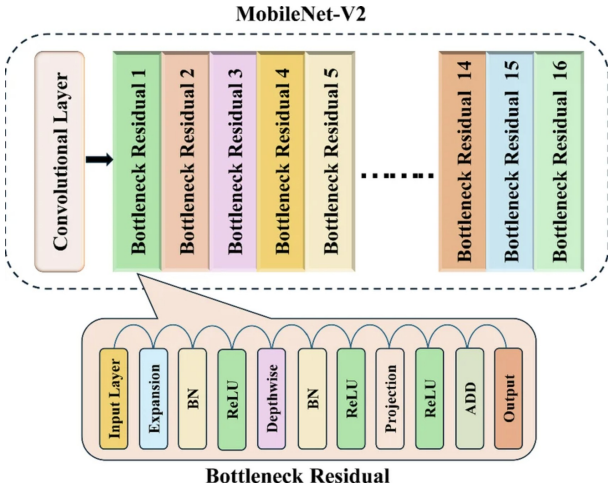


Fig. 12. Architecture of the MobileNetV2 network, illustrating the core convolutional layer, the sequence of bottleneck residual blocks, and the detailed structure of an individual bottleneck residual block including expansion, depthwise convolution, and projection layers [1]

For medical imaging tasks such as polyp characterization, MobileNetV2 is often preferred over heavier architectures like ResNet, DenseNet, or EfficientNet because it offers a favorable balance between accuracy and efficiency. Its lightweight design minimizes overfitting on small datasets and enables faster training and inference, making it ideal for environments with limited GPU resources while still delivering reliable diagnostic performance.

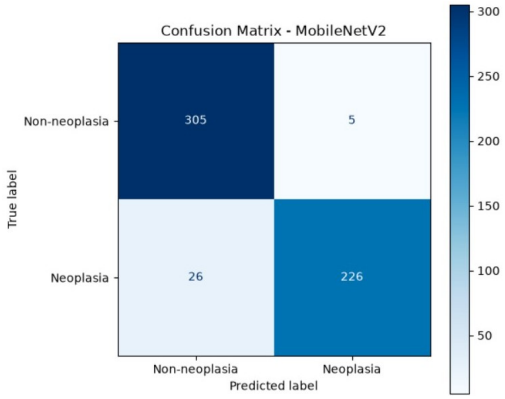


Fig. 13. Confusion matrix for the MobileNetV2 classifier

Metric	Value
Accuracy	0.9448
Sensitivity	0.8968
Specificity	0.9839
F1 Score	0.9358
AUC	0.9849

TABLE XII
PERFORMANCE METRICS FOR THE PRETRAINED MOBILENETV2 MODEL

Across all evaluation outputs, the model demonstrates consistently strong discriminative ability, with the ROC curve (AUC = 0.985) indicating excellent separation between neoplasia and non-neoplasia. In clinical applications, metrics that do not include true negatives in the numerator, namely Sensitivity, Precision (PPV), F1-score, and AUC,are the most meaningful, because they reflect the model's ability to correctly identify disease and avoid harmful false reassurance. Here, the model achieves high sensitivity (0.90) and exceptionally high precision (0.98), meaning it detects the majority of neoplastic cases while rarely producing false alarms. The F1-score (0.94) further confirms balanced performance in identifying true neoplasia, and the confusion matrix shows a low false-positive burden but a moderate number of false negatives (n=26), which is the most clinically consequential error type.

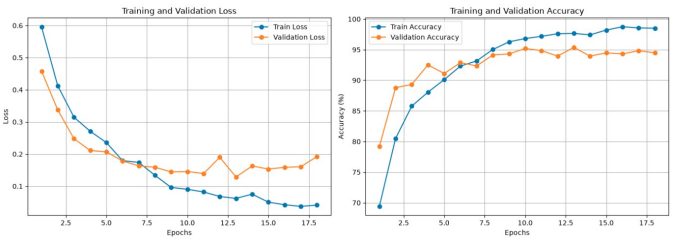


Fig. 14. Training and validation loss (left) and accuracy (right) curves over 18 epochs for the MobileNetV2 model

The training and validation curves suggest mild overfitting: training accuracy approaches 98–99%, while validation accuracy stabilizes around 94–95%, and validation loss becomes noisier after approximately ten epochs. This pattern indicates that the model has learned some training-specific patterns but still generalizes reasonably well. Importantly, the validation performance remains stable and high, implying that overfitting is present but not severe enough to undermine internal validity. Nevertheless, in a clinical context, even mild overfitting warrants caution, as performance may degrade when applied to external datasets from different scanners, institutions, or patient populations.



Fig. 15. Qualitative examples of polyp characterization results, displaying input CCE images alongside their true and predicted diagnostic labels (neoplasia vs. non-neoplasia)

All things considered, the model exhibits clinically promising performance, particularly in metrics that matter for patient safety and diagnostic reliability. Its high sensitivity, precision, F1-score, and AUC suggest strong potential for aiding neoplasia detection. However, the presence of false negatives and mild overfitting means that external validation and threshold optimization are essential before considering the model clinically safe.

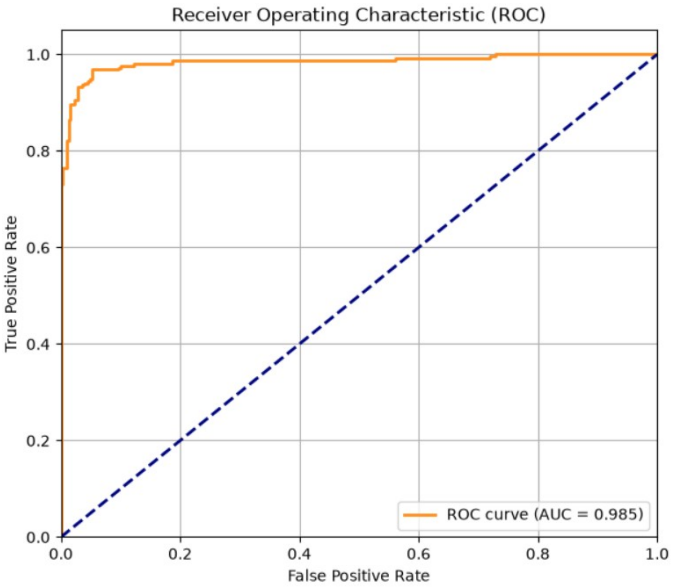


Fig. 16. ROC curve for the pretrained MobileNetV2, achieving an AUC of 0.985

Across the models, the most clinically important metrics are Sensitivity, Precision/PPV, F1-score, and AUC, because these do not include true negatives in the numerator and therefore better reflect the model's ability to correctly identify neoplasia without providing false reassurance. Qualitatively, MobileNetV2 and DenseNet121 appear to offer the strongest balance of clinically meaningful performance, combining high sensitivity with strong precision and excellent discriminative ability. ResNet18 shows comparatively higher sensitivity, suggesting it may miss fewer neoplastic cases, but its trade-off is slightly weaker precision and overall balance.

Member	Model	Pretrained
Samamidou	MobileNetV2	Yes
Bikiaris	ResNet18	No
Drzyzgiewicz	EfficientNet	Yes
Rozier	DenseNet121	Yes

TABLE XIII
TEAM MEMBERS AND MODEL CONFIGURATIONS FOR POLYP CHARACTERIZATION

Model	ACC	SENS	SPEC	F1	AUC
MobileNetV2	0.9502	0.9127	0.9806	0.9426	0.9895
ResNet18	0.9431	0.9524	0.9355	0.9375	0.9846
EfficientNet	0.8210	0.9020	0.6750	0.8670	0.9020
DenseNet121	0.9649	0.9414	0.9817	0.9578	0.9898

TABLE XIV
PERFORMANCE COMPARISON ACROSS TEAM MEMBERS (DUMMY VALUES).

B. Semantic Segmentation

TransUNet is a hybrid medical-image segmentation architecture that combines the strengths of convolutional neural networks with transformer-based self-attention. It retains the familiar encoder–decoder structure used widely in clinical imaging, but enhances the encoder with Vision Transformers (ViTs) to capture long-range contextual dependencies that standard convolutions often miss. The CNN encoder extracts low-level spatial features, while the transformer layers model global relationships across the entire image—an important capability for medical tasks where anatomical structures and pathological patterns may span large spatial regions.

In the decoder, TransUNet uses U-Net–style skip connections to fuse high-resolution convolutional features with the globally contextualized transformer representations, producing segmentation maps that are both structurally precise and contextually informed. This hybrid design allows TransUNet to outperform purely convolutional models in scenarios where fine boundary delineation and global anatomical consistency are critical, such as tumor segmentation, organ localization, and lesion mapping. Its ability to integrate local detail with global context makes it particularly well suited for clinical applications requiring high reliability and interpretability.

The confusion matrix indicates that the model correctly identifies a very large number of polyp pixels (286,411 true

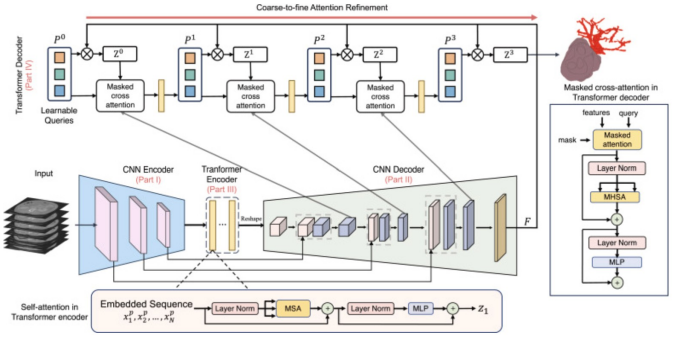


Fig. 17. Overview of the TransUNet architecture combining a Vision Transformer (ViT) encoder with a U-Net decoder via skip connections for medical image segmentation [2]

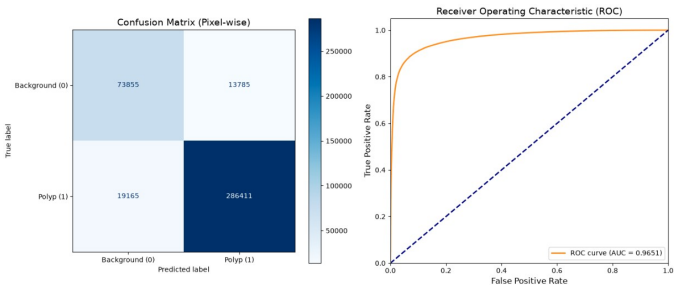


Fig. 18. Confusion maxtric and ROC curve for TransUNet polyp segmentation, demonstrating an AUC of 0.9651.

positives), which is essential because missed polyp tissue can lead to under-diagnosis or incomplete polyp removal. The relatively high number of false negatives (19,165 polyp pixels predicted as background) is more clinically concerning: in practice, these missed regions could correspond to subtle polyp borders or flat lesions that are harder to detect but clinically important. Meanwhile, the false positives (13,785 background pixels predicted as polyp) are less dangerous clinically—over-segmentation may lead to slightly exaggerated polyp boundaries, but it does not risk missing pathology. The ROC curve, with an AUC of 0.9651, shows that the model has strong discriminative ability overall. In clinical terms, this means the system is highly capable of distinguishing polyp tissue from normal mucosa across a wide range of thresholds. A high AUC is particularly important in medical imaging because segmentation models are often used as decision-support tools, therefore reliability is a requirement.

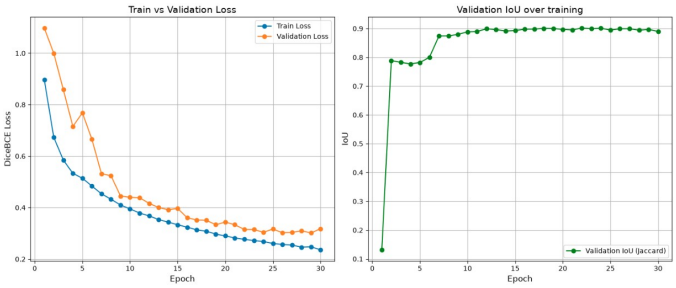


Fig. 19. Training and validation loss curves (left) alongside the IoU curve (right) across training epochs for TransUNet

These training curves reinforce the earlier interpretation and show a model that is learning stable, clinically meaningful segmentation behaviour. The loss curves demonstrate a consistent decrease in both training and validation loss, with no divergence between them. This indicates that the model is not overfitting and is generalizing well to unseen data—an essential property in clinical imaging, where variability in lighting, mucosal texture, and polyp morphology can be substantial. The fact that validation loss remains slightly higher than training loss is expected and suggests the model is not memorizing the training set but learning transferable features.

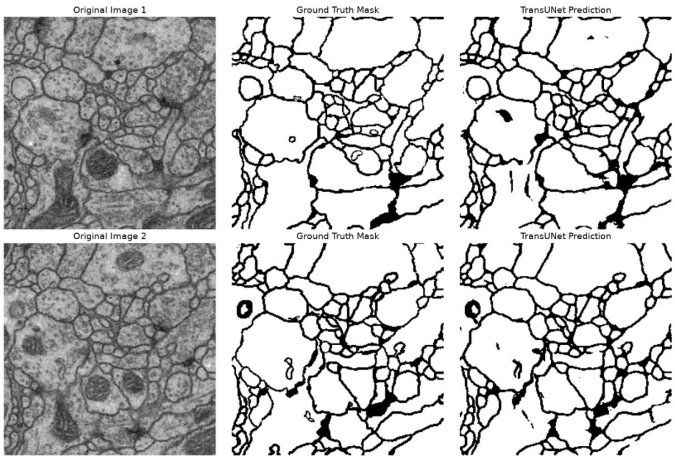


Fig. 20. TransUNet semantic segmentation examples

The IoU curve shows a rapid rise in segmentation quality during the early epochs, followed by a plateau near high IoU values. Clinically, this is significant: IoU reflects how accurately the model captures the true polyp boundaries. Stability near 0.9 indicates that the model consistently delineates polyp regions with high precision. The early stabilization also suggests that the chosen architecture and loss function provide strong gradient signals, enabling the model to quickly learn the structural cues that differentiate polyp tissue from surrounding mucosa.

Member	Model	Pretrained
Samamidou	Trans-UNet	Yes
Bikiaris	UNet	Yes
Drzyzgiewicz	DeepLabV3	Yes
Rozier	Attention U-Net	Yes

TABLE XV
TEAM MEMBERS AND MODEL CONFIGURATIONS FOR SEMANTIC SEGMENTATION

Model	ACC	PREC	RECC	F1	IoU
Trans-UNet	0.9162	0.9541	0.9373	0.9456	0.8969
UNet	0.9159	0.9454	0.9464	0.9456	0.8974
DeepLabV3	0.7520	0.4230	0.3480	0.3810	0.236
Attention-UNet	0.9050	0.9300	0.9400	0.9350	0.8850

TABLE XVI
PERFORMANCE COMPARISON ACROSS TEAM MEMBERS

TransUNet and UNet show very similar performance profiles, with both achieving a balanced combination of high recall and high precision, resulting in stable F1-scores and solid IoU values. TransUNet's transformer-enhanced encoder suggests slightly stronger global context modeling, while UNet maintains robust local feature extraction, making both well suited for polyp segmentation. Attention-UNet demonstrates good recall, but slightly lower precision and IoU, indicating more false detections and less accurate boundary delineation.

III. Detecting Epilecptic seizures in EEG

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures arising from abnormal, hypersynchronous neuronal activity. Electroencephalography (EEG) is the primary clinical tool for detecting these dynamics, offering fast temporal resolution essential for assessing seizure patterns and supporting differential diagnosis. Its ability to capture transient, non-visible biomarkers makes EEG indispensable for both routine evaluation and advanced computational approaches to seizure detection in time-series analysis.

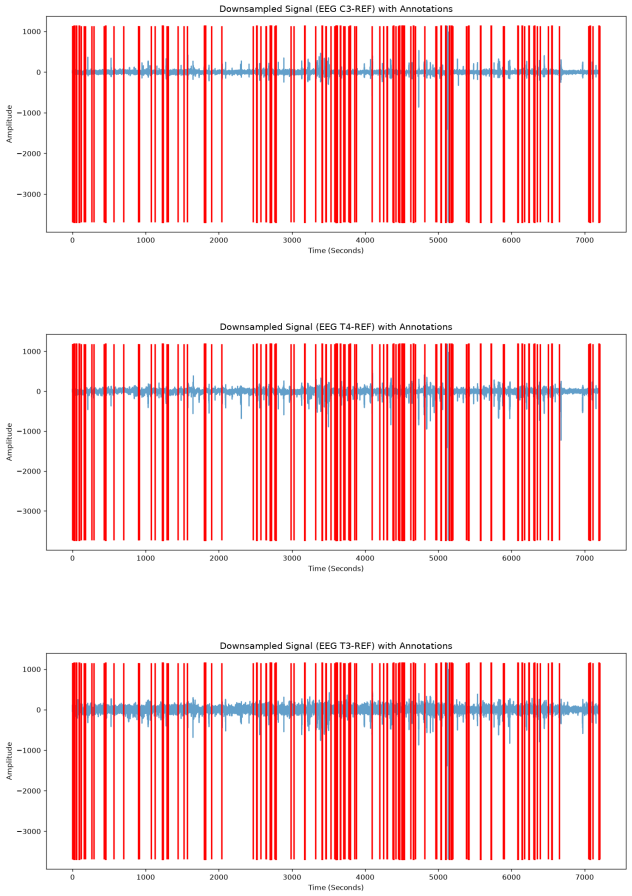


Fig. 21. Representative surface-level EEG time-series recording capturing absence seizure activity (channels 4,5,13, annotations in red)

This work employs a surface-level EEG dataset consisting of approximately two-hour recordings capturing clinically verified absence seizures, providing continuous time-series data suitable for a machine-learning pipeline.

A. Data Description

EEG is well-established in the literature as a non-stationary signal, meaning its statistical properties (mean, variance, spectral content) vary over time due to changes in brain state, vigilance, cognitive load, and transitions into and out of epileptic activity [3][4]. A clear example comes from studies of absence epilepsy, where spike-and-wave discharges (SWDs) produce abrupt, structured oscillatory patterns that differ markedly from interictal background activity, demonstrating strong deviations from stationarity . This non-stationarity is also evident in analyses of dynamic propagation and time-frequency evolution of absence-seizure EEG, which show rapid shifts in spectral power and synchronization across cortical regions

Knowing whether an EEG time series is stationary is crucial from a methodological perspective, considering that many signal-processing and statistical tools (e.g., AR models, certain spectral estimators, classical hypothesis tests) assume stationarity, and violating this assumption can lead to biased parameter estimates, misleading spectral characterizations, and degraded classifier performance. For seizure-detection pipelines, explicitly acknowledging non-stationarity guides the choice of appropriate preprocessing and motivates the use of time-resolved or state-dependent models.

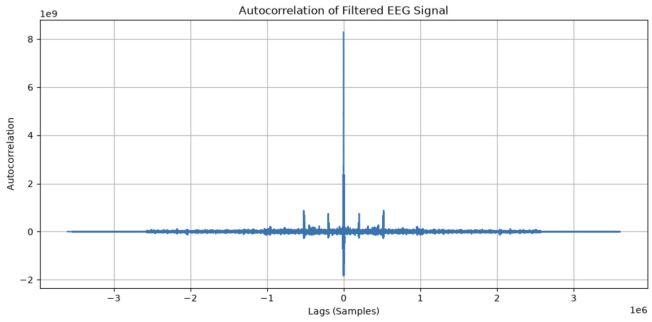


Fig. 22. Autocorrelation plot of the filtered EEG signal calculated across multiple time lags

After filtering the signal with the bandpass filter described further in section III.B, its autocorrelation is calculated and plotted. From an end-to-end perspective, the autocorrelation resembles that of white noise, therefore one would be led to assume stationarity. An implementation of the Augmented Dickey-Fuller test confirms this assumption, returning a p-value of virtually zero. However, zooming into the region around zero reveals a decaying behavior of the autocorrelation, which from this perspective resembles that of a random walk, which makes it non-stationary. This is the final conclusion, considering that it is apparent from the plot in the appropriate scale, and that it is well-established in the relative literature.

B. Data Processing

To suppress low-frequency drift and high-frequency noise, the raw EEG signal is filtered using a linear-phase Finite Impulse Response (FIR) band-pass filter. The filter is designed

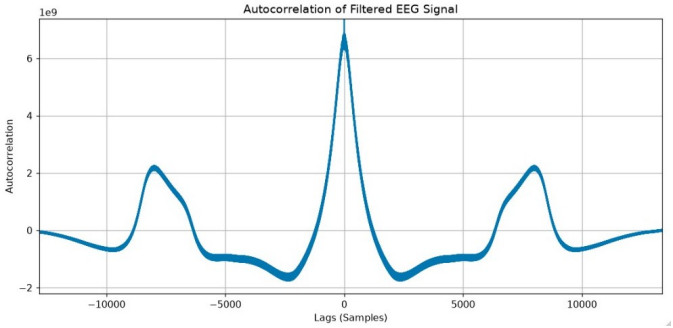


Fig. 23. Autocorrelation plot of the filtered EEG signal across lags surrounding zero

Statistic	Value
ADF Statistic	-5.3279
p-value	4.7924e-06

TABLE XVII
AUGMENTED DICKEY–FULLER (ADF) TEST RESULTS.

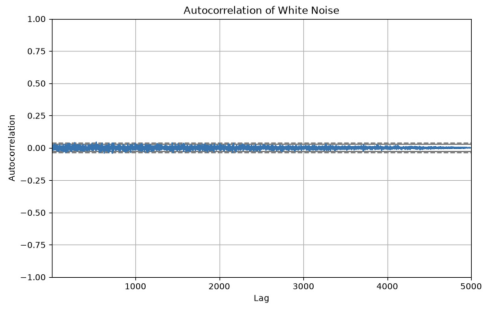


Fig. 24. Autocorrelation of a white noise implementation

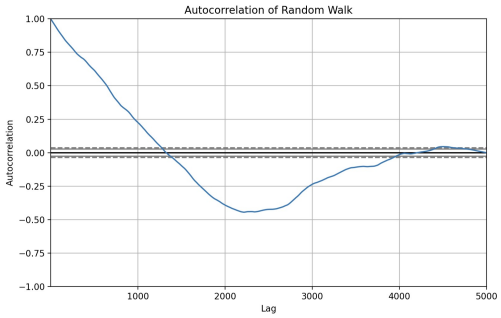


Fig. 25. Autocorrelation of a random walk implementation

using a passband of 1–70 Hz and a total length of 101 taps. This configuration effectively removes DC offsets, slow movement artifacts, and muscle or electrical interference while preserving the physiologically relevant EEG rhythms. The filtering operation is applied through convolution, ensuring that the temporal structure of the signal remains

The pipeline additionally includes an optional normalization step in which each EEG channel is scaled by its maximum absolute amplitude. This procedure standardizes the dynamic range across channels, improves numerical stability, and reduces inter-channel variability. Although normalization is disabled in the current configuration, the mechanism is available for scenarios where amplitude differences between channels may bias downstream feature extraction or classification.

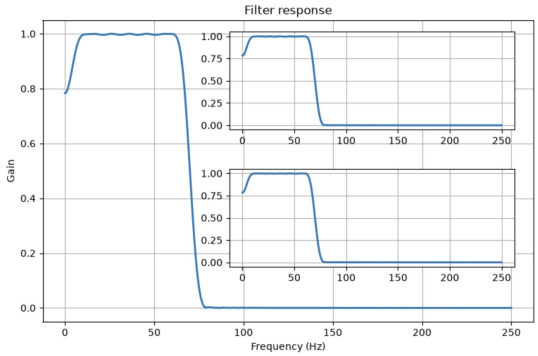


Fig. 26. FIR bandpass filter response in the frequency domain

The EDF file contains clinical annotations marking seizure events. These annotations are extracted from the file header and stored as intervals defined by their onset time and duration. This information is later used to label each time window according to its degree of overlap with seizure periods. The annotation extraction step enables the construction of supervised learning datasets and supports the evaluation of seizure-related signal characteristics.

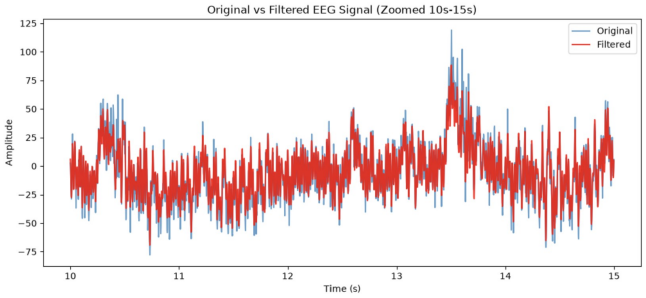


Fig. 27. Filtered vs unfiltered signal comparison

The filtered EEG is segmented into overlapping windows to facilitate localized feature extraction. Each window spans 3 s of data with a 1 s overlap, producing a sequence of partially overlapping segments that capture the temporal evolution of the signal. For each window, the pipeline records the window's center time, the corresponding signal segment, and the proportion of overlap with annotated seizure intervals. Between a rectangular and a Hamming window, the rectangular window proved itself more efficient in this particular classification task. This segmentation strategy transforms the continuous EEG into a structured set of labeled segments suitable for machine learning analysis.

As for the feature extraction, for every window, a 14-dimensional feature vector is computed, combining statistical descriptors, non-linear measures, and spectral characteris-

tics. Time-domain features capture amplitude distribution and signal complexity, while frequency-domain features quantify rhythmic activity across canonical EEG bands. Spectral estimates are obtained using Welch's method, ensuring robust power spectral density (PSD) estimation even for short windows.

Let $x[n]$ denote the EEG samples within a window and let $\mathrm{PSD}(f)$ denote the power spectral density estimated via Welch's method. The feature vector consists of the following components:

- **Mean:**

$$\mu = \frac{1}{N}\sum_{n=1}^{N} x[n]$$

- **Median:**

$$\mathrm{median}(x)$$

- **Variance:**

$$\sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x[n]-\mu)^2$$

- **Kurtosis:**

$$\kappa = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{x[n]-\mu}{\sigma}\right)^4$$

- **Linear Energy:**

$$E_{\mathrm{lin}} = \sum_{n=1}^{N} x[n]^2$$

- **Non-linear Energy (Teager–Kaiser):**

$$E_{\mathrm{TKEO}} = \frac{1}{N-2}\sum_{n=2}^{N-1}\big(x[n]^2 - x[n-1]\,x[n+1]\big)$$

- **Shannon Entropy:**

$$H = -\sum_{i=1}^{K} p_i \log_2 p_i$$

where p_i are the histogram-based probability estimates of the signal amplitude.

- **Band Powers:**

$$P_\delta = \sum_{f\in[0.5,4]} \mathrm{PSD}(f), \quad P_\theta = \sum_{f\in(4,8]}, \quad P_\alpha = \sum_{f\in(8,13]},$$

$$P_\beta = \sum_{f\in(13,30]}, \quad P_\gamma = \sum_{f>30}$$

- **Peak Frequency:**

$$f_{\mathrm{peak}} = \arg\max_f \mathrm{PSD}(f)$$

- **Spectral Edge Frequency (SEF 90%):**

$$f_{\mathrm{SEF90}} = \min\left\{ f : \frac{\sum_{f'\le f}\mathrm{PSD}(f')}{\sum_{f'}\mathrm{PSD}(f')} \ge 0.9 \right\}$$

The concatenation of these features across all channels yields a structured feature matrix of dimension

$$\text{num_windows} \times (\text{num_channels} \cdot 14),$$

which forms the basis for downstream classification, clustering, or visualization tasks.

To prepare the data for machine learning, the feature matrix is divided into training and validation subsets. Two strategies are supported: a balanced split, in which seizure and non-seizure windows are sampled proportionally, and a simple chronological split based on a predefined ratio. In the current configuration, the chronological split is used, assigning the first 80% of windows to the training set and the remaining 20% to the validation set. Each window is assigned a binary label indicating whether it overlaps with a seizure interval. This step transforms the continuous EEG recording into a supervised learning dataset suitable for classification tasks.

Before dimensionality reduction or model training, the feature vectors undergo numerical cleaning to eliminate invalid values, which are replaced with zeros. Subsequently, the features are standardized using the StandardScaler, which centers each feature to zero mean and scales it to unit variance. Standardization is essential for ensuring that all features contribute equally to downstream learning algorithms, preventing dominance by high-amplitude or high-variance components. To reduce linear dependencies among features and improve computational efficiency, Principal Component Analysis (PCA) is applied to the standardized training data. PCA is configured to retain 90% of the total variance, ensuring that the most informative components are preserved while redundant dimensions are removed. Whitening is enabled to decorrelate the principal components and equalize their variances. The resulting transformation is applied consistently to both training and validation sets. This step yields a compact, orthogonal representation of the EEG feature space, facilitating more stable and efficient model training.

C. Seizure Detection

Two supervised learning models were evaluated for the task of seizure detection from EEG windows: Logistic Regression and a Support Vector Machine (SVM) with an RBF kernel. Logistic Regression was selected as a baseline linear classifier due to its interpretability and stability in high-dimensional feature spaces. The SVM was chosen as a more flexible non-linear model capable of capturing complex decision boundaries inherent in EEG data. It should be mentionned that the SVM was optimized using GridSearchCV, with the explicit goal of minimizing false negatives. Therefore, the hyperparameter search prioritized configurations that reduce the likelihood of

misclassifying seizure windows as normal activity, even at the cost of slightly increasing false positives.

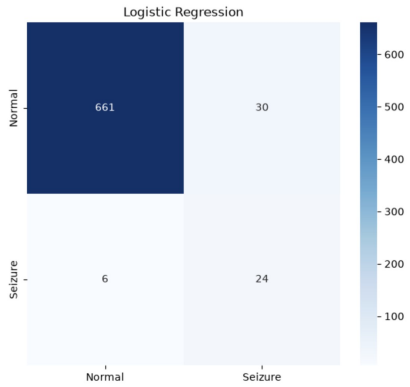


Fig. 28. Confusion matrix evaluating the performance of the Logistic Regression model on the EEG seizure detection task

The confusion matrix for Logistic Regression demonstrates strong overall performance, with 661 true negatives and 24 true positives, indicating that the model reliably distinguishes normal from seizure activity. The number of false positives (30) is acceptable in clinical screening contexts, where over-alerting is preferable to under-detecting. However, the presence of 6 false negatives is clinically relevant, since these represent seizure windows that were incorrectly classified as normal. The ROC curve further supports this conclusion, with an AUC of 0.941, indicating excellent discriminative ability. The curve rises steeply toward the upper-left corner, reflecting high sensitivity across a wide range of thresholds. This is particularly important in clinical monitoring systems, where sensitivity is often prioritized over specificity.

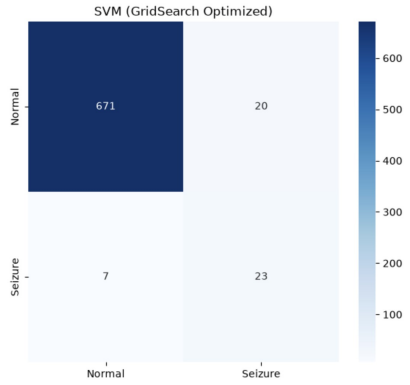


Fig. 29. Confusion matrix for SVM optimized via GridSearchCV for minimizing false negatives

The SVM optimized via GridSearch shows slightly improved raw classification counts compared to Logistic Regression: 671 true negatives and 23 true positives, with 20 false positives and 7 false negatives. The reduction in false positives aligns with the model's ability to better separate normal EEG patterns from seizure-related abnormalities. However, the presence of 7 false negatives, despite the optimization strategy, demonstrates the inherent difficulty of detecting subtle seizure activity in short EEG windows. The ROC curve for the optimized SVM shows an AUC of 0.932, slightly lower than Logistic Regression but still indicative of strong performance. The curve remains well above the diagonal random baseline, confirming that the SVM is a reliable classifier. The small difference in AUC between the two models suggests that both approaches capture the essential structure of the EEG feature space, though Logistic Regression may generalize slightly better.

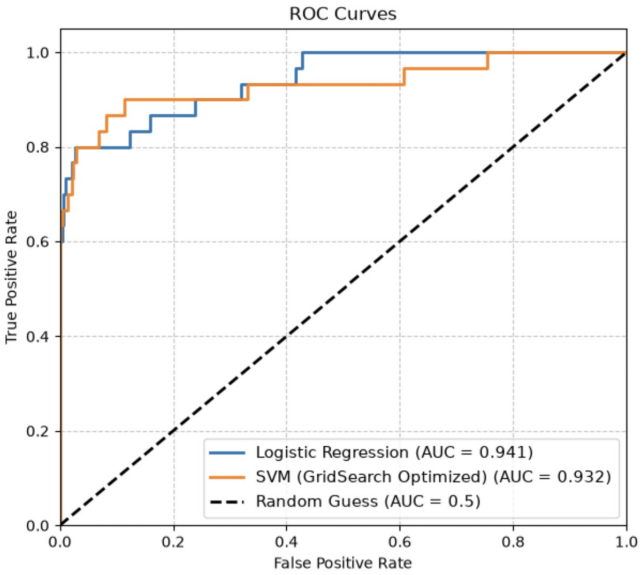


Fig. 30. ROC curve for the Logistic Regression seizure detector, achieving an AUC of 0.941

Given that false negatives are clinically unacceptable, Logistic Regression may be preferable in seizure detection pipelines where sensitivity is most important. The SVM's lower false-positive rate may be beneficial in settings where alarm fatigue is a concern, but only if seizure detection is not compromised.

IV. Conclusion and Discussion

This report has comprehensively examined the various application of different fields of artificial intelligence in the medical field. Through the evaluation of deep learning architectures such as MobileNetV2 and TransUNet for polyp characterization and segmentation, alongside machine learning classifiers for EEG seizure detection, the efficacy of computer-aided diagnosis in modern clinical pipelines has been thoroughly demonstrated.

Despite these encouraging findings, several limitations of the project and of our current implementations should be noted. The models were primarily evaluated on standardized benchmark datasets, which may not fully capture the complex real-world variability, noise profiles, and artifact distributions encountered in live clinical practice. Additionally, time constraints restricted the depth of hyperparameter optimization, while reliance on standard feature extraction approaches without extensive fine-tuning may limit model adaptability across diverse patient populations and acquisition hardware.

While the current methodology provides robust performance across benchmark datasets, the analysis of Section III could very much be extended using more advanced mathematical and signal processing techniques—such as higher-order statistics, short-time Fourier transform (STFT), and wavelets—but was not pursued due to time constraints. Incorporating these advanced feature extraction methods in future work holds significant potential to capture non-stationary signal dynamics more precisely and further elevate diagnostic accuracy in clinical environments.

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